

SETUP AND DEPLOYMENT INSTRUCTIONS

The installation process is divided into two distinct phases:

Phase 1: Tailscale Configuration

This phase involves generating necessary authentication keys and gathering required information for the virtual machine installation. Key activities include:

- * Generating Tailscale authentication keys.
- * Creating API access tokens for uninstall functionality.
- * Collecting the Tailnet DNS name information.

Phase 2: Virtual Machine Installation

This phase consists of two components:

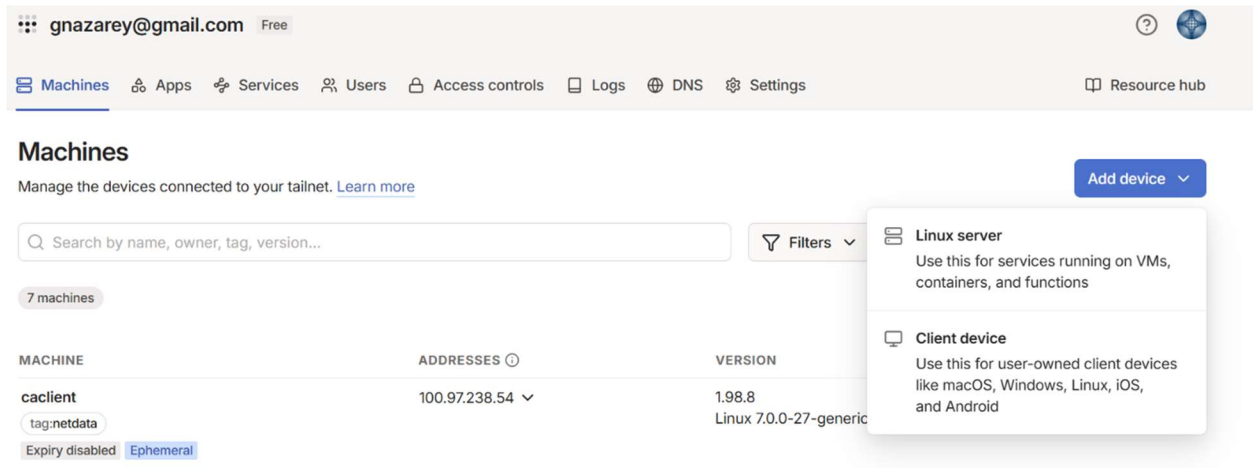
- * Parent Server Installation (Primary server setup).
- * Client Server Installations (Multiple client installations).

Both phases must be completed sequentially for successful deployment.

TAILSCALE CONFIGURATION STEPS

1. Generate an Auth Key

(This key will be copied to the configuration file during the Parent and Client server setup.)



- Log in to your Tailscale account.
- Click on the "Add Device" pull-down menu on the right side of the screen.
- Select "Linux Server."

Add Linux server

Select preferences for your server to generate an install script.

1. Set up device

Tags



Authenticate and control access to the device using a tag. Tagging a device disables node key expiry for that device. [Learn more](#)

[Manage tags in Access Controls](#)

Ephemeral



Automatically remove the device from your tailnet when it goes offline. [Learn more](#)

Use as exit node



Route traffic through this device on your Tailscale network. [Learn more](#)

d. Under *Set up device*: Set Tags (On) and fill in a tag name; set Ephemeral (On); set Use as exit node (Off).

2. Set up authentication key

Reusable



Use this key to authenticate more than one device. [Learn more](#)

 Use this as part of automation to set up multiple devices.

Auth key expiration

Set a time period to authenticate devices with this auth key. This will not affect the node key expiry of any device authenticated with this auth key. [Learn more](#)

 — + days

Must be between 1 and 90 days.

 Set the key expiry no longer than needed, to reduce impact if it is lost. To continuously add devices to your tailnet, such as with third-party integrations, [use an OAuth client](#).

e. Under *Set up authentication key*: Set Reusable (On) and Expires in: 7 days.

f. Click "Generate install script."

g. Copy the resulting script to Notepad.

2. Generate an API Key

(This is needed for the uninstall process.)

MachinesAppsServicesUsersAccess controlsLogsDNSSettingsResource hub

Tailnet Settings

General

User management

Device management

Policy file management

Trust credentials

Webhooks

Contact preferences

Billing

Personal Settings

Keys

Keys

View and manage your Auth keys and API access tokens.

Your private device keys are not included here: they are always private, stay on your device, and are never shared with Tailscale. [Learn more](#)

Auth keys

Authenticate devices without an interactive login. [Learn more](#)

Generate auth key...

ID	CREATED	EXPIRY	TYPE
> k6rMu9R7xy11CNTRL	Jul 6, 2026	Jul 13, 2026	Reusable, Ephemeral Revoke...
> khCqmJ6qJc11CNTRL	Jul 7, 2026	Jul 14, 2026	Reusable, Ephemeral Revoke...

> 1 recently invalidated auth key

API access tokens

Access tokens give access to the Tailscale API. [Learn more](#)

Generate access token...

ID	CREATED	EXPIRY	TYPE
> kvbWm4ALN11CNTRL	Jul 6, 2026	Oct 4, 2026	API access token Revoke...

- Log in to your Tailscale account.
- Click on "Settings" on the right of the top menu.
- Click on "Keys" under the Personal Setting menu item on the left.

Generate API access token ×

Description
Add an optional description for the key.

Netplan

Expiration
Number of days this key is valid for.

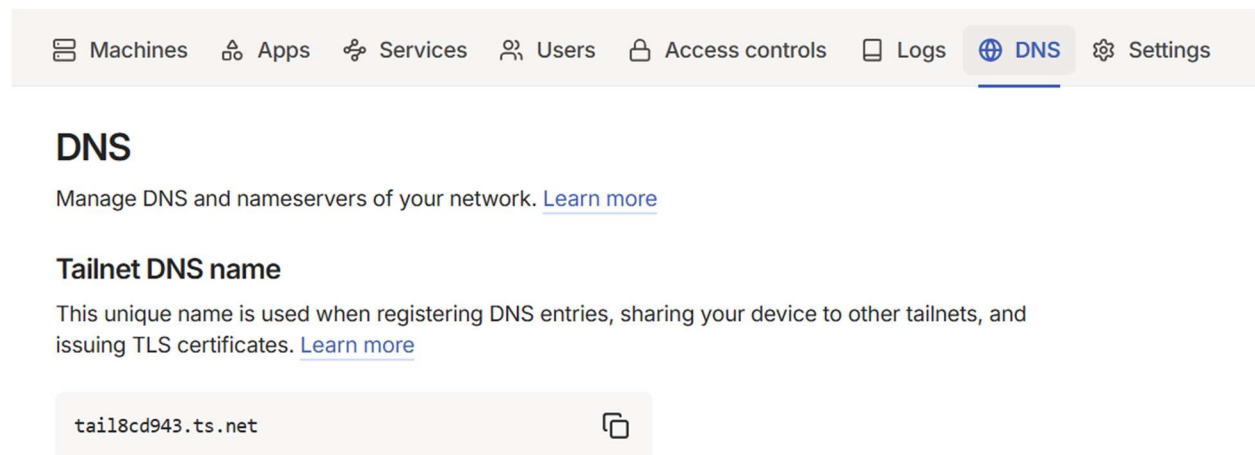
90 — + days

Must be between 1 and 90 days.

Cancel Generate access token

- d. Click on "Generate access token," located to the right of the API access tokens header in the middle of the screen.
- e. Provide a description for the access token and keep the default 90 days under Expiration.
- f. Click "Generate access token."
- g. Copy the generated token to Notepad.

3. Gather the Tailnet DNS Name



- a. Log in to your Tailscale account.
- b. Click on "DNS" on the top menu.
- c. The first item listed under DNS is the *Tailnet DNS Name*. Copy this name to Notepad.

EDITING THE server_config.sh FILE

The default content of `server\_config.sh` is:

```
#####
# User Configuration
#####
HOSTNAME="INSERT_HOST_NAME_HERE"
TAILSCALE_AUTH_KEY="INSERT_TAILSCALE_AUTH_KEY_HERE"
# Only need these if you wish to use the uninstall functionality
TAILSCALE_API_KEY="INSERT_TAILSCALE_API_KEY_HERE"
TAILNET="INSERT_TAILNET_HERE"

#####
# Script Generated
#####
API_KEY="YOUR_GENERATED_UUID_HERE"
PARENT_IP="A.B.C.D"
```

Configuration Notes:

1. HOSTNAME must be unique for each host.

2. TAILSCALE_AUTH_KEY: Take the key from "Generate a Auth Key." The format is typically ``tskey-auth-0123456789ABCDEFGH-IJKLMNOPQRSTUVWXYZ0123456789abc`` . Include the entire key, including the prefix ``tskey-auth-`` .

3. TAILSCALE_API_KEY: Take the key from "Generate an API Key." The format is typically ``tskey-api-0123456789ABCDEFGH-IJKLMNOPQRSTUVWXYZ0123456789abc`` . Include the entire key, including the prefix ``tskey-api-`` .

4. TAILNET: This is the Tailnet DNS Name obtained from "Gather the Tailnet DNS Name."

VIRTUAL MACHINE INSTALLATION

Use the image file: [Lab Image Ubuntu 26 04 LTS.ova](#)

PARENT CONFIGURATION (Parent Server Setup)

New virtual machine

✓ 1 Select creation type

2 Select OVF and VMDK files

3 Select storage

4 License agreements

5 Deployment options

6 Additional settings

7 Ready to complete

vmware

Select creation type

How would you like to create a Virtual Machine?

Create a new virtual machine

Deploy a virtual machine from an OVF or OVA file

Register an existing virtual machine

This option guides you through the process of creating a virtual machine from an OVF and VMDK files.

Back

Next

Finish

Cancel

1. Selection Creation Type: Deploy a virtual machine from an OVF or OVA File. Click Next.

The screenshot shows the 'New virtual machine - parent_server' window. On the left is a sidebar with a list of steps: 1 Select creation type, 2 Select OVF and VMDK files (highlighted), 3 Select storage, 4 License agreements, 5 Deployment options, 6 Additional settings, and 7 Ready to complete. The main area is titled 'Select OVF and VMDK files' and contains the instruction 'Select the OVF and VMDK files or OVA for the VM you would like to deploy'. Below this is a text input field for the VM name, which contains 'parent_server'. A note states: 'Virtual machine names can contain up to 80 characters and they must be unique within each ESXi instance.' A large light blue rectangular area is intended for file selection. At the bottom of this area, a file named 'Lab_Image_Ubuntu_26_04_LTS.ova' is listed with a small 'vm' icon. The VMware logo is in the bottom left corner. At the bottom right are four buttons: 'Back', 'Next', 'Finish', and 'Cancel'.

New virtual machine - parent_server

✓ 1 Select creation type
2 Select OVF and VMDK files
3 Select storage
4 License agreements
5 Deployment options
6 Additional settings
7 Ready to complete

Select OVF and VMDK files

Select the OVF and VMDK files or OVA for the VM you would like to deploy

Enter a name for the virtual machine.

parent_server

Virtual machine names can contain up to 80 characters and they must be unique within each ESXi instance.

×  Lab_Image_Ubuntu_26_04_LTS.ova

vmware

Back Next Finish Cancel

2. Select OVF and VMDK files: Name the virtual machine "parent_server." Select and copy the OVA to the ESXi host. Click Next.

New virtual machine - lab_server

- ✓ 1 Select creation type
- ✓ 2 Select OVF and VMDK files
- ✓ 3 Select storage
- 4 License agreements
- 5 Deployment options
- 6 Additional settings
- 7 Ready to complete

Select storage

Select the storage type and datastore

☒ Standard ☐ Persistent Memory

Select a datastore for the virtual machine's configuration files and all of its virtual disks.

Name	Capacity	Free	Type	Thin pro...	Access
datastore1	20.75 GB	19.34 GB	VMFS6	Supported	Single
NAS	96 TB	16.39 TB	NFS	Supported	Single
Nvme	1.69 TB	771.14 GB	VMFS6	Supported	Single

3 items

Back Next Finish Cancel

3. Select Storage: Select a datastore with at least 60GB free space. Click Next.

New virtual machine - lab_server

- ✓ 1 Select creation type
- ✓ 2 Select OVF and VMDK files
- ✓ 3 Select storage
- ✓ 4 **Deployment options**
- 5 Ready to complete

Deployment options

Select deployment options

Network mappings	Server Server
Disk provisioning	☒ Thin ☐ Thick
Power on automatically	☐

Back Next Finish Cancel

4. Deployment options: Select a network mapping that has internet access; select Thin disk provisioning; UNCHECK Power on Automatically. Click Next.

New virtual machine - parent_server

- ✓ 1 Select creation type
- ✓ 2 Select OVF and VMDK files
- ✓ 3 Select storage
- ✓ 4 Deployment options
- 5 **Ready to complete**

A required disk image was missing.

Review your settings selection before finishing the wizard

Product	Lab_Image_Ubuntu_26_04_LTS
VM Name	parent_server
Files	Lab_Image_Ubuntu_26_04_LTS-disk1.vmdk
Datastore	Nvme
Provisioning type	Thin
Network mappings	Server: Server
Guest OS Name	Unknown

! Do not refresh your browser while this VM is being deployed.

Back Next Finish Cancel

5. Ready to complete: Review the configuration (ignore any error message that pop-up). Click Finish.

CLIENT CONFIGURATION (Client Server Setup)

1. Selection Creation Type: Deploy a virtual machine from an OVF or OVA File. Click Next.

The screenshot shows the 'New virtual machine - client01' wizard in VMware vSphere. The left sidebar contains a progress list with five steps: 1. Select creation type (checked), 2. Select OVF and VMDK files (highlighted in blue), 3. Select storage (checked), 4. Deployment options (checked), and 5. Ready to complete (checked). The main panel is titled 'Select OVF and VMDK files' and includes the instruction 'Select the OVF and VMDK files or OVA for the VM you would like to deploy'. Below this, there is a text input field for the virtual machine name, which contains 'client01'. A note states: 'Virtual machine names can contain up to 80 characters and they must be unique within each ESXi instance.' A large light blue rectangular area is intended for selecting files. Inside this area, a file named 'Lab_Image_Ubuntu_26_04_LTS.ova' is listed with a small 'vm' icon to its left. At the bottom right of the window, there are four buttons: 'Back', 'Next', 'Finish', and 'Cancel'.

2. Select OVF and VMDK files: Name the virtual machine "client01." Select and copy the OVA to the ESXi host. Click Next.

New virtual machine - lab_server

- ✓ 1 Select creation type
- ✓ 2 Select OVF and VMDK files
- ✓ 3 Select storage
- 4 License agreements
- 5 Deployment options
- 6 Additional settings
- 7 Ready to complete

Select storage

Select the storage type and datastore

Standard Persistent Memory

Select a datastore for the virtual machine's configuration files and all of its virtual disks.

Name	Capacity	Free	Type	Thin pro...	Access
datastore1	20.75 GB	19.34 GB	VMFS6	Supported	Single
NAS	96 TB	16.39 TB	NFS	Supported	Single
Nvme	1.69 TB	771.14 GB	VMFS6	Supported	Single

3 items

Back Next Finish Cancel

3. Select Storage: Select a datastore with at least 60GB free space. Click Next.

New virtual machine - lab_server

- ✓ 1 Select creation type
- ✓ 2 Select OVF and VMDK files
- ✓ 3 Select storage
- ✓ 4 Deployment options
- 5 Ready to complete

Deployment options

Select deployment options

Network mappings Server Server

Disk provisioning ☒ Thin ☐ Thick

Power on automatically ☐

Back Next Finish Cancel

4. Deployment options: Select a network mapping that has internet access; select Thin disk provisioning; UNCHECK Power on Automatically. Click Next.

New virtual machine - client01

- ✓ 1 Select creation type
- ✓ 2 Select OVF and VMDK files
- ✓ 3 Select storage
- ✓ 4 Deployment options
- 5 Ready to complete**

A required disk image was missing.

Review your settings selection before finishing the wizard

Product	Lab_Image_Ubuntu_26_04_LTS
VM Name	client01
Files	Lab_Image_Ubuntu_26_04_LTS-disk1.vmdk
Datastore	Nvme
Provisioning type	Thin
Network mappings	Server: Dev
Guest OS Name	Unknown

! Do not refresh your browser while this VM is being deployed.

Back Next Finish Cancel

5. Ready to complete: Review the configuration (ignore any error message that pop-up). Click Finish.

Repeat these steps for each required remote data location.

User Accounts Information:

Username: labuser

Password: DontT311@ny0n3 - This user has sudo privileges.

VM BOOT AND CONFIGURATION

Power On Instructions:

Once all virtual machines have completed their installation process, power them on.

Network Information Gathering:

VM Tools are pre-installed on each image, which enables network information collection after boot-up and DHCP address assignment. Each VM will automatically obtain its network configuration upon startup.

Remote Access:

SSH is enabled on all images, allowing remote access to each VM using its assigned IP address for continued configuration tasks.

Recommended Workflow:

1. Power on all virtual machines.
2. Wait for system boot and network initialization.
3. Collect IP addresses from VM Tools or DHCP leases.
4. Establish SSH connections using the assigned IP addresses.
5. Proceed with additional required configuration steps.

Installation Files (Links):

* Configuration File:

https://raw.githubusercontent.com/gnazarey/tailscale_exercise/refs/heads/main/server_config.sh

* Installation File:

https://raw.githubusercontent.com/gnazarey/tailscale_exercise/refs/heads/main/install.sh

Lab Parent Configuration Steps:

1. Log in as `labuser` with the given credentials.
2. In the user's home directory, run these commands:


```
/usr/bin/wget  
https://raw.githubusercontent.com/gnazarey/tailscale_exercise/refs/heads/main/server_config.sh
```

```
/usr/bin/wget  
https://raw.githubusercontent.com/gnazarey/tailscale_exercise/refs/heads/main/install.sh
```

3. Edit `server_config.sh`:

- * Change HOSTNAME to 'server'.
- * Change TAILSCALE_AUTH_KEY to the key you created.

4. Run this command: `bash install.sh -u` (The '-u' flag indicates parent/updater).

5. When the script finishes, make sure to copy the `client_config.sh` file to all client machines.

Lab Client Configuration Steps:

1. Log in as `labuser` with the given credentials.

2. In the user's home directory, run this command:

```
/usr/bin/wget  
https://raw.githubusercontent.com/gnazarey/tailscale_exercise/refs/heads/main/install.sh
```

3. Copy the `client_config.sh` file or create it using the output from the server script in the home directory.

4. Edit `client_config.sh`:

- * Change HOSTNAME to 'client'.
- * Verify that TAILSCALE_AUTH_KEY, Parent_IP, and API_KEY match the values found in the `server_config.sh` file on the parent machine.

5. Run this command: `bash install.sh -c` (The '-c' flag indicates client).